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COMPUTER ENGINEERING DEPARTMENT

Lecture 7: Probabilistic (or Randomized) Algorithms

Monte Carlo algoritmasından bahsedilecek

Example 1: Dataset Comparison

$$X = x_1x_2 \dots x_n, \quad x_i \in \{0,1\}$$

$$Y = y_1y_2 \dots y_n, \quad y_i \in \{0,1\}$$



Question: $X \stackrel{?}{=} Y$

Classical Communication Protocol:

Comparing the two datasets bit by bit:

- Communication overhead: $O(n)$
- Comparison overhead: $O(n) \Rightarrow n = 10^{16} \text{ bits } (B) \approx 1136 \text{ TB}$

burdaki tüm örnekler polinom zamanda çözülebilir örnekler, ama bu yöntemler np problemleri de çözmeye yarıyor

normal yöntemle çözersek gönderme (communication) ve karlatırma karmaklı $O(n)$

azaltmak için randomize algorithm kullanacaz

$v(X) = \text{value}(X) = \sum_{i=1}^n x_i \cdot 2^{n-i}$
 bit serisinin decimal (onluk) değerini veren bir fonksiyon
 $\Pi(k) = \{p \in \mathbb{P} \mid p \leq k\}, \quad \pi(k) = |\Pi(k)|$
 k'ya esit ya da k'dan küçük olan tüm asal sayılar kumesi
 kumenin eleman sayisi
 Randomized Equal algorithm:
 Input: $X = x_1x_2 \dots x_n, Y = y_1y_2 \dots y_n, x_iy_i \in \{0,1\}$
 A: 101101
 B: 101101
 6 haneli oldugu ucun 36
 bit sayi

1. A randomly chooses a prime number $p \in \Pi(n^2)$
2. A calculates the "fingerprint of $X \Rightarrow s = v(X) \bmod p$.
 a. **Note:** X is considered as a binary number
3. A sends s and p to B
4. B calculates the "fingerprint of $Y \Rightarrow t = v(Y) \bmod p$
5. B compares whether $s = t$? Are the two fingerprints the same?
 a. Yes: equal $\rightarrow A$,
 b. No: unequal $\rightarrow A$

0 ile 36 arasında bir prime number bulalım, mesela 29 $p = 29$

X sayisinin decimalini bulup prime numberla modluyoruz

$32+8+4+1 = 45 \quad 45\%29 = 16 \quad s = 16$
 şimdi s ve p değerlerimizi Bye gönderiyoruz.

B'nin bitlerinin decimal hali ve A'nin seçtiği p sayısı ile mod işlemi tekrar yapıyor. Cevap s'ye esitse, esit değilse esit değil kabul ediliyor

Comparison overhead:

The Equal algorithm is a randomized communication protocol for the data comparison example. It drastically reduces the comparison effort. Comparison overhead was previously $O(n)$, now the overhead is reduced to 5 steps $\rightarrow O(1)$



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Communication overhead:

$$0 \leq p, s \leq n^2$$

$$m \in \mathbb{N} \rightarrow \lceil \log m \rceil \text{ Bits}$$

$$l(s, p) \leq 2 * \lceil \log n^2 \rceil \leq 4 * \lceil \log n \rceil$$

For $n = 10^{16}$;

$$\leq 4 * 16 * \lceil \log 10 \rceil$$

$$\leq 4 * 16 * 4$$

$$= 256 \text{ Bits}$$

Instead of 10^{16} , we would only transmit 256 bits.

$$Prob_{equal}[A | B]$$

$$X = Y \Rightarrow t = s$$

$$X \neq Y \Rightarrow t \neq s$$

$$Prob_{equal}["unequal" | X = Y] = 0$$

Example:

$$n = 5$$

$$X = 10011 \quad v(X) = 19$$

$$Y = 10001 \quad v(Y) = 17$$

$$\text{for } p = 11 \Rightarrow s = 8, t = 6$$

$$X = 10011 \quad v(X) = 19$$

$$Y = 10001 \quad v(Y) = 17$$

$$\text{for } p = 2 \Rightarrow s = 1, t = 1$$

$p = 2$ is a bad witness for the inequality of X and Y

Question: How many bad witnesses are in $\Pi(n^2)$?

$s = t$, i.e. $v(X) = v(Y) \bmod (p)$, although $X \neq Y$.

$$\Pi^+(n^2, X, Y) = \{p \in \mathbb{P} \mid v(X) \neq v(Y)(p), X \neq Y\}$$

$$\Pi^-(n^2, X, Y) = \{p \in \mathbb{P} \mid v(X) = v(Y)(p), X \neq Y\}$$



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Establishing the relationship between bad witnesses and total witness candidates:

$$Prob_{equal}[X = Y | X \neq Y] = \frac{|\Pi^-(n^2, X, Y)|}{|\Pi(n^2)|} = \frac{2 \ln n}{n}$$

ispatı gösterilmedi,
sonuc budur

$$\text{for } n = 10^{16} \Rightarrow \frac{2 \ln 10^{16}}{10^{16}} \approx 0.7 * 10^{-14}$$

cok dusuk bir bad witness
orani var, dolayisiyla iyidir.

Multiple rounds of algorithm to further reduce the error:

$$l = 10 \Rightarrow 0.7 * 10^{-144}$$

her calistirildiginda yanlis cikma ihtimali
daha da dusuyor

$$O(l * \log n)$$

$$L \subseteq \Sigma^*, w \in \Sigma^*: w \in^? L$$

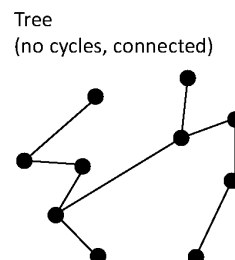
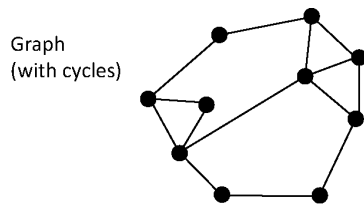
$$D = \{(x, y) \in \{0,1\}^n \times \{0,1\}^n \mid n \in \mathbb{N}_0, x \neq y\}$$

Given $w \in \{0,1\}^n \times \{0,1\}^n$

1. $Prob_{equal}["w \in D" \mid w \notin D] = 0$
2. $Prob_{equal}["w \in D" \mid w \in D] \geq 1 - \frac{2 \ln n}{n} \Rightarrow RP(\varepsilon(n)), \varepsilon(n) \text{ is the error bound}$
3. Overhead: $O(\log n)$

Example 2: Triangle Graph

$$\Delta Graph = \{ \langle G \rangle \mid G \text{ undirected graph that contains at least one triangle} \}$$





Randomized Triangle Graph Algorithm:

1. T (tester) randomly chooses an edge $\{a, b\}$
2. T randomly chooses a node $c \neq \{a, b\}$
3. Test: Do c and $\{a, b\}$ form a triangle? $\rightarrow$ Yes or No

$$Prob_T["yes" | G \notin \Delta Graph] = 0 \rightarrow \text{no false positive statement}$$

$$G = (V, E)$$

$$|V| = n, |E| = m$$

$$Prob_T["yes in one round" | G \in \Delta Graph] \geq \frac{3}{m} * \frac{1}{n-2}$$

$$Prob_T["no in one round" | G \in \Delta Graph] \leq 1 - \frac{3}{m(n-2)}$$

$$Prob_T["no in l rounds" | G \in \Delta Graph] \leq \left(1 - \frac{3}{m(n-2)}\right)^l$$

$$Prob_T["yes in l rounds" | G \in \Delta Graph] \geq 1 - \left(1 - \frac{3}{m(n-2)}\right)^l$$

$$\left(1 + \frac{1}{k}\right)^k \xrightarrow{k \rightarrow \infty} e^1 = 2.7182 \dots$$

More general formulation with x instead of the special case with 1:

$$\left(1 + \frac{x}{k}\right)^k \xrightarrow{k \rightarrow \infty} e^x$$

Place a minus before x:

$$\left(1 + \frac{-x}{k}\right)^k \xrightarrow{k \rightarrow \infty} e^{-x}$$

We want to apply the Euler sequence for our probabilistic $\Delta Graph$ algorithm:

$$k = l, x = yl \rightarrow (1 - y)^l \xrightarrow{l \rightarrow \infty} e^{-yl}$$

$$y = \frac{3}{m(n-2)}, l = \frac{m(n-2)}{3} \rightarrow y * l = 1$$

$$\left(1 - \frac{3}{m(n-2)}\right)^{\frac{m(n-2)}{3}} \approx e^{-1} \approx \frac{1}{e} \approx \frac{1}{2.7} < \frac{1}{2}$$



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RP

1. $Prob_T["yes" \mid G \notin \Delta Graph] = 0$
2. $Prob_T["yes" \mid G \in \Delta Graph] \geq \frac{1}{2}$
3. *polynomiell*

Example 3: Probabilistic Prime Number Tests

Little Fermat's Theorem:

$$\begin{aligned} p \in \mathbb{P}, a \in \mathbb{N}, (a, p) &= 1 \\ \Downarrow \\ a^{p-1} &= 1(p) \end{aligned}$$

Question: $n \in \mathbb{N}, a \in \mathbb{N}$ with $a^{n-1} = 1(n) \Rightarrow n \in ? \mathbb{P}$

$$\begin{aligned} 2^{n-1} &= 1(n) \\ n &= 3, 4, 5 \dots \end{aligned}$$

$n=341$ – the number is not prime, but it pretends to be prime
 $\Rightarrow$ Smallest pseudoprime number to base 2

Definition: Let m be a composite number with $(a, m) = 1$ and $a^{m-1} = 1(m)$ or $a^m = a(m)$, then m is called pseudoprime to base a

$n=341$ – the number is not prime, but it pretends to be prime

- $n=341$ is smallest pseudoprime number to base 2
- $n=341$ is not a pseudoprime to base 3

In other words: Pseudoprime numbers satisfy Fermat's little theorem even though they are not prime

Existence of composite numbers that are pseudoprime to all (coprime) bases:

A composite number is called a Carmichael number iff $a^{m-1} = 1(m)$
or $a^m = 1(m)$ applies to all bases with $(a, m) = 1$

- $m=561$ is the smallest Carmichael number
- Carmichael numbers are free of squares
- Factoring Carmichael numbers contains at least 3 different prime factors
- There are infinitely many Carmichael numbers: 561, 1105, 1729, 2465, 2821, ...



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Theorem: For $m \in \mathbb{N}, m \geq 3$, let

$$F_m = \{a \in \mathbb{Z}_m \mid a^{m-1} = 1(m)\}$$

be the set of bases for which m passes the Fermat test.

If m is not a prime number, then F_m contains the bases that “fool” the Fermat test.

Let $m \in \mathbb{N}, m \geq 3$, be a composite and not a Carmichael number, then the following applies:

$$|F_m| \leq \frac{\mathbb{Z}_m}{2}$$

alg notPrime ($k \in U_+, k \geq 3$)

Randomly pick an $a \in (1, \dots, k-1)$ with $(a, k) = 1$

If $a^{k-1} \neq 1(k)$
 then Output: *k is not prime*
 otherwise Output: *k prime?*

endalg

$\overline{\mathbb{P}} = \text{COMPOSITES} \in \text{RPP}$

(1) $\text{Prob}[k \in \overline{\mathbb{P}} \mid k \notin \overline{\mathbb{P}}] = 0$

(2) $\text{Prob}[k \notin \overline{\mathbb{P}} \mid k \in \overline{\mathbb{P}}] \leq \frac{1}{2}$

(3) notPrime is polynomial

The probability of error is therefore at most $\frac{1}{2}$. If the algorithm is now carried out for l rounds in which the base a is chosen anew at random and independently, then the probability of error is at most $\frac{1}{2^l}$; so, it can be made as small as you want.

➔ Executing the algorithm l times leads to an error probability of $\leq \frac{1}{2^l}$



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Miller Rabin Algorithm and b-Sequences

$\mathbb{Z}_m$ is a field iff $m \in \mathbb{P}$

$$x^2 = 1$$

$\mathbb{Z}_m: x = 1, x = -1 \Rightarrow \text{trivial solutions}$

$x \in \mathbb{Z}_m$ is called a non-trivial square root of 1 modulo m if $x^2 = 1$ and $x \neq 1$ and $x \neq -1$.

e.g., $m = 15: x = 4, x = -4 \Rightarrow \text{non-trivial solutions}$

In $\mathbb{Z}_m$, x^2 has the solution $x = \pm 1$ iff $m \in \mathbb{P}$

In other words: If there is a nontrivial square root modulo m, then m is a composite number.

$$m \in \mathbb{N}$$

$$s = \max\{r \in \mathbb{N} \mid 2^r \mid m-1\}$$

$$d = \frac{m-1}{2^s}$$

$b \in \mathbb{Z}_m: b\text{-sequence}$

$$\langle b^{2^0 d}, b^{2^1 d}, b^{2^2 d}, \dots, b^{2^{s-1} d}, b^{2^s d} \rangle \text{ mod } (m)$$

If $m \in \mathbb{P}$:

$$b^{2^s d} = b^{2^s \cdot \frac{m-1}{2^s}} = b^{m-1} = 1(m)$$

➔ We get more information about the structure of the b-sequence if m is a prime number:

Examples:

a) Let $m=25$ (i.e., composite), then $s=3$ and $d=3$. For the basis $b=2$ the sequence results

$$\langle 2^3, 2^6, 2^{12}, 2^{24} \rangle = \langle 8, 14, 21, 16 \rangle$$

For $b=3$ we get

$$\langle 3^3, 3^6, 3^{12}, 3^{24} \rangle = \langle 2, 4, 16, 6 \rangle$$



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and for $b = 7$

$$\langle 7^3, 7^6, 7^{12}, 7^{24} \rangle = \langle 18, -1, 1, 1 \rangle$$

b) Let $m=97$ (i.e., prime), then $s=5$ and $d=3$. For the basis $b=2$ the sequence results

$$\langle 2^3, 2^6, 2^{12}, 2^{24}, 2^{48}, 2^{96} \rangle = \langle 8, 64, 22, -1, 1, 1 \rangle$$

For $b=14$ this results

$$\langle 14^3, 14^6, 14^{12}, 14^{24}, 14^{48}, 14^{96} \rangle = \langle 28, 8, 64, 22, -1, 1 \rangle$$

for $b = 35$

$$\langle 35^3, 35^6, 35^{12}, 35^{24}, 35^{48}, 35^{96} \rangle = \langle 1, 1, 1, 1, 1, 1 \rangle$$

and for $b = 62$

$$\langle 62^3, 62^6, 62^{12}, 62^{24}, 62^{48}, 62^{96} \rangle = \langle -1, 1, 1, 1, 1, 1 \rangle$$

➔ The examples show that the b -sequences for prime numbers have a specific structure.

Theorem: Let $b \in \mathbb{P}$, $s = \max\{r \mid 2^r \mid p-1\}$, $d = \frac{p-1}{2^s}$, $b \in \mathbb{N}$ with $(b, p) = 1$

Then (1) $b^d = 1 \pmod{p}$ or

$$(2) \exists r \in \{0, 1, \dots, s-1\}: b^{2^r d} = -1 \pmod{p}$$

If $p \in \mathbb{P}$, then the b -sequence has one of the following forms:

$$(1) \langle 1, 1, \dots, 1 \rangle$$

$$(2) \langle -1, 1, \dots, 1 \rangle$$

$$(3) \langle ?, ?, \dots, ?, -1, 1, \dots, 1 \rangle$$

Last element of the b -sequence: $b^{p-1} = 1 \pmod{p}$



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Use the reverse of the b-sequence:

$m \in \mathbb{N}, b \in \mathbb{Z}_m$, with b – sequence

(1) $\langle ?, ?, \dots, 1, 1, \dots, 1 \rangle$

(2) $\langle ?, ?, \dots, ?, -1 \rangle$

(3) $\langle ?, ?, \dots, ?, ? \rangle$

$\rightarrow m \notin \mathbb{P}$

Definition: $m \in U_+, m \geq 3, m - 1 = 2^s d$ with $d \in U_+, b \in \mathbb{Z}_m$

If $b^d = 1(m)$ or $b^{2^r d} = -1(m)$ holds for an $r \in \{0, 1, \dots, s - 1\}$, then m is called **strong pseudoprime** to base b .

Theorem: $m \in U_+, m \geq 3$, composite, then the number of bases for which m is **strongly pseudoprime** is at most $\frac{m-1}{4}$

$\rightarrow$ Miller Rabin Algorithm

algorithm MILLER-RABIN($n \in U_+, n \geq 3$)

Compute d and s with $n - 1 = d \cdot 2^s$ and d positive uneven

Randomly pick an $a \in \{2, 3, \dots, n - 2\}$

$b := a^d(n)$

if $b = 1(n)$ or $b = -1(n)$: Output: n is prime?

for $r := 1$ to $s - 1$ do

$b := b^2(n)$

if $b = -1(n)$: Output: n is prime?

if $b = 1(n)$: Output: n is not prime

endfor

Output: n is prime?

endalgorithm MILLER-RABIN



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Miller Rabin Algorithm $\in$ RPP

- (1) $Prob[Output: n \text{ is not prime} \mid n \in \mathbb{P}] = 0$
- (2) $Prob[Output: n \text{ is prime?} \mid n \notin \mathbb{P}] \leq \frac{1}{4}$
- (3) $Prob[l - \text{times Output: } n \text{ is prime?} \mid n \notin \mathbb{P}] \leq \frac{1}{4^l}$
- (4) $Prob[\text{after } l \text{ executions Output: } n \text{ is not prime} \mid n \notin \mathbb{P}] \geq 1 - \frac{1}{4^l}$
- (5) $O(l * (\log n))$ arithmetical operations or $O(l * (\log n)^3)$ bit operations

Outline:

$Dataset \in RP\left(\frac{2 \ln n}{n}\right)$

$\Delta Graph \in RP\left(\frac{1}{2}\right)$

$Miller Rabin \in RP\left(\frac{1}{4}\right)$

Complexity Class RP:

Random/Probabilistic polynomial running algorithms with one-sided error

- No false positive statements.
- Conversely, RP algorithms make errors with a particular bound ϵ .
- By repeating the execution of the algorithm, the total error can be reduced.

$\Rightarrow$ The algorithms that class RP defines are also called **Monte Carlo algorithms**. Such algorithms allow a one-sided error.